Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	The vector sum of the momenta of bodies in a system stays constant (even if forces act between the bodies), ... (1) ....provided there is no external / resultant force / in an isolated system (1) Accept: The total momentum before a collision is equal to the total momentum after a collision....... (1) ....provided there is no external / resultant forces act / in an isolated system (1)	2			2		
		(ii)	Momentum before collision = 5.4×10^3 (N s) (1) [from graph] Momentum after collision = (5 000 or ans to *(b)*(ii)+ 7 000) v (1) v = 0.45 (m s^{-1}) (1) [**ecf** on value from graph, including slips in power of 10]	1	1 1		3	3	
			Question 5 total	**5**	**5**	**0**	**10**	**6**	**0**